

Christopher M. Heggerud

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Academic Appointments

2025-current Assistant Professor - Department of Mathematics, University of Manitoba.

2021-2025 Postdoctoral Researcher - Department of Environmental Science & Policy, University of California, Davis. Hastings Lab (NSERC PDF, 2023 - 2025).

Education

2021 PhD in Applied Mathematics, University of Alberta.
Thesis titled "Modelling phytoplankton across many scales: transient dynamics, human interactions, and niche differentiation in the light spectrum", supervised by Dr. Mark Lewis and Dr. Hao Wang.

2015 Bachelor of Science, Toronto Metropolitan University.
(Formerly Ryerson University) Bachelor of Science with honors in Applied Mathematics with a minor in Medical Physics. Supervised by Dr. Kunquan Lan.

Research Experience

2025- current University of Manitoba

Assistant Professor in the department of Mathematics specializing in Mathematical Biology.

2021-2025 University of California, Davis [Postdoctoral Research]

Mentored by Dr. Alan Hastings at University of California, Davis, Department of Environmental Science and Policy.

2015-2021 University of Alberta [Research Assistant]

Mentored by Dr. Mark Lewis and Dr. Hao Wang at the University of Alberta, Department of Mathematical and Statistical Sciences.

2014 NSERC [Undergraduate Student Research Award]

Supervised by Dr. Kunquan Lan at Ryerson University, Department of Mathematics.

2013 NSERC [Undergraduate Student Research Award]

Supervised by Prof. Dejan Delic at Ryerson University, Department of Mathematics.

Publications

For publications with 8 or more authors only the first author is listed.

16. X. Lui, et al. Experimental warming amplifies temporal variance scaling across diverse soil microbiota, In review (2025)
15. K. Feng, et al. The Power-Law of Species-Time Relationships in Microbial Ecology, In review (2025)

14. J. Serpico, B.A. Zambrano-Luna, R. Milne, **C. M. Heggerud**, A. Hastings, H. Wang. Deciphering culprits for cyanobacterial blooms and lake vulnerability in north-temperate lakes, *Ecological Modeling* In Press (2026)
13. A. Morozov, U. Feudel, A. Hastings, K.C. Abbott, K. Cuddington, **C. M. Heggerud**, S. Petrovskii, Long-living transients in ecological models: recent progress, new challenges, and open questions. *Physics of Life Reviews*, 51 (2024)
12. **C. M. Heggerud** & A. Hastings, Predicting transient dynamics through the use of empirical dynamical modelling: A case study of anaerobic digestion. *Journal of the Royal Society Interface*, 21:20240059 (2024)
11. K.C. Abbot, **C.M. Heggerud**, YC. Lai, A. Morozov, S. Petrovski, K. Cuddington, & A. Hastings. When and why ecological systems respond to the rate rather than the magnitude of environmental changes. *Biological Conservation*, 292:110494 (2024)
10. **C. M. Heggerud**, J. Xu, Hao Wang, M. A. Lewis, R. Zurawell, C. Loewen, R. Vinebrooke, & P. Ramazi, Predicting imminent cyanobacterial blooms in lakes using incomplete timely data. *Water Resources Research* 60, 2 (2024)
9. **C. M. Heggerud**, KY. Lam & H. Wang. Niche differentiation in the light spectrum promotes coexistence of phytoplankton species: a spatial modelling approach. *Journal of Mathematical Biology* 86, 54 (2023)
8. A. Shen, S. Gao, **C. M. Heggerud**, H. Wang, Z. Ma, & S. Yuan. Fluctuation of growth and photosynthetic characteristics in *Prorocentrum shikokuense* under phosphorus limitation: Evidence from field and laboratory. *Ecological Modelling*, 479, (2023).
7. **C. M. Heggerud**, K.C. Abbott, A. Hastings, Transient Dynamics. *Oxford Bibliographies in Ecology*. New York: Oxford University Press, 2023.
6. V. Kirkow, H. Wang, P. V. Garcia, S. Ahmed, **C. M. Heggerud**, Impacts of a changing environment on a stoichiometric producer-grazer system: a stochastic modelling approach. *Ecological Modelling* 469 (2022).
5. H. Wang, P. V. Garcia, S. Ahmed, **C. M. Heggerud***, Mathematical comparison and empirical review of the Monod and Droop forms for resource-based population dynamics. *Ecological Modelling* 466 (2022). (*Supervisory Author)
4. **C. M. Heggerud**, H. Wang, & M. A. Lewis, "Coupling the socio-economic and ecological dynamics of cyanobacteria: Single lake and network dynamics." *Ecological Economics* 194 (2022).
3. A. Peace, et al. Stoichiometric Ecotoxicology for a Multisubstance World. *BioScience*: biaa160, (2021).
2. **C. M. Heggerud**, H. Wang, & M. A. Lewis, Transient dynamics of a stoichiometric cyanobacteria model via multiple-scale analysis. *SIAM Journal of Applied Mathematics* 80 (3), 1223–1246 (2020).
1. **C. M. Heggerud**, & K.Q. Lan. Local stability analysis of ratio-dependent predator-prey models with predator harvesting rates. *Applied Mathematics and Computation* 270: 349-357 (2015).

Works in progress

- a. **C. M. Heggerud**, & A. Hastings. Detecting transient crawl-by dynamics and their

behaviour.

- b. J. Serpico, B.A. Zambrano-Luna, R. Milne, **C. M. Heggerud**, A. Hastings, H. Wang. Coupled human-lake dynamics reveal two-way feedback between land cover and cyanobacterial blooms.
- c. J. Kaare-Rasmussen, **C. M. Heggerud**, & S. Schreiber. Tipping points in Forest Ecosystems

Student Mentorship

^P PhD, ^M MSc, ^U Undergraduate, * informal mentorship

2026- current Rahul Nivash ^U

Physics undergraduate researching chaotic dynamics at the intersection of biology and physical systems. Awarded the FoS USRA at U Manitoba.

2025- current Marzieh Roshani ^M (Mathematics)

A current MSc in Mathematics at the University of Manitoba under my sole supervision. Tentative thesis title: "Detecting rate induced tipping in human-environmental systems"

2023- current Jakob Kaare-Rasmussen ^{P*}(Population Biology)

Informal mentor of Jakob for research and career during their PhD studies at UC Davis.

2023- current Jennifer Paige ^{P*} (Applied Mathematics)

Informal mentor of Jennifer for research and career during their PhD studies at UC Davis.

2026 Alec MacIver ^U

A statistics undergraduate researching various tipping points in a noisy system modelling plant-fungal interaction. Part of the MXML course.

2021 Velizar Kirkow ^U (Mathematics)

Co-supervised Velizar Kirkow from University of Exeter who was visiting with a MITACS Globalink Research Internship.

2019 Daniel Jin ^U (Biology)

Co-supervised Daniel Jin for their biology undergraduate research project (BIOL498) at the University of Alberta.

2019 Julian Pavón García ^U (Mathematics)

Co-supervised Julian Pavón García from Eberhard Karls University of Tübingen who was visiting with a MITACS Globalink Research Internship.

Teaching Experience

2025- current University of Manitoba

As an Assistant Professor I have taught: MATH 1510-Applied Calculus I (F25), MATH 1220-Linear Algebra (Honors)(W26), Manitoba Experiential Math Learning (Undergraduate research course)(W26), and graduate reading courses.

2024 University of California, Davis [Instructor]

COSMOS - Mathematical Modeling of Biological Systems – An intensive four-week summer residential school for high school students with high aptitude for careers in STEM.

2022-2023 University of California, Davis [Primary Instructor]

MAT 21A (Differential Calculus)(F22).

Mathematical Biology Seminar (Co-organizer and instructor on record W2023)

2015–2021 University of Alberta [Teacher's Assistant]
2014-2015 Ryerson University [Teacher's Assistant]
2014-2015 The Math Guru [Tutor]

Funding/Awards/Fellowships

*Major

2026-2031 NSERC Discovery Grant*
Held at the University of Manitoba. (Major Award- \$190K)
2026-2031 NSERC Launch Supplement*
Held at the University of Manitoba. (Major Award- \$12.5K)
2025-2030 University of Manitoba Start-Up Grant*
Held at the University of Manitoba. (Major Award)
2023-2025 NSERC postdoctoral fellowship*
Held at UC Davis, supervised by Alan Hastings. (Major Award)
2021 Doctoral Dissertation Award
University of Alberta. Awarded to top 5% of theses in the Faculty of Science.
2020 Alberta Graduate Excellence Scholarship*
University of Alberta (Major Award)
2019 Society of Mathematical Biology - Travel Grant
University of Alberta
2019 Faculty of Graduate Studies and Research - Travel Award
University of Alberta
2017/2018 Alberta Innovates - Graduate Student Scholarship*
Held at the University of Alberta. (Major Award)
2015/2016 NSERC [Canadian Graduate Scholarship - Masters]*
Held at the University of Alberta. (Major Award)
2015 Walter H. Johns Graduate Fellowship
University of Alberta

Academic & Professional Activities

2025- current Lead organizer – Models in Population Dynamics, Ecology and Evolution 2027.
Conference to take place in Winnipeg, June 2027. 2027 (~100 attendees)
2025- current Lead organizer – Banff International Research Station (BIRS) 5-day workshop. Titled
“From transient dynamics to tipping points: Understanding qualitative changes in biological
systems” to take place in W27.
2025- current Lead organizer and re-establisher, U Manitoba Mathbio Group
2025- current Organizer, Maud Menton Institute seminar series at the University of Manitoba
2023- current Member of the advisory committee for the Population Dynamics, Ecology, and
Evolution subgroup of the Society of Mathematical Biology.
2023-2025 Postdoc representative, UC Davis Department of Environmental Science and Policy
Diversity, Equity and Inclusion Committee
2024-2025 PNAS Journal Club Panelist.
2024 Mini Symposium Organizer, “Biological Dynamics Across the Many Natural Scales”.
SIAM life science 2024.

2024	Special Session co-organizer, “Dynamical Systems Modeling for Biological and Social Systems”. Joint Mathematics Meeting 2024.
2023/24	Co-organizer of the Dynamics Days US meeting
2022-2025	Co-organizer for the UC Davis Mathematical Biology Seminar.
2023	Mini Symposium co-organizer, “Microbial and ecological dynamics across the many natural scales”. Society of Mathematical Biology meeting annual meeting.
2021-2025	Postdoctoral Associate, Center for Population Biology, University of California, Davis.
2021	Banff International Research Station (BIRS) Workshop: Mathematics of Human Environmental Systems. Presenter and active participant.
2019	Mini Symposium co-organizer, “Resource explicit population models”. Society of Mathematical Biology meeting annual meeting.
2018	Investigative Workshop on Stoichiometric Ecotoxicology [NIMBioS- Knoxville Tennessee]

Academic Talks

*Invited Talks, ^oOnline, ^cContributed Talks, ^LLectures

Summary: 1 keynote talk, 7 invited department seminar talks, 1 industry/government seminar talk, 18 invited research conference talks, 1 invited education talk, 2 guest lectures.

2026	The different ways ecological change can happen and how systems respond* Invited conference talk at the AMS spring central meeting.
2026	Predicting and understanding transient dynamics in ecological systems* ^o Invited research talk at the Workshop on Transient Dynamics: Theory and Applications hosted by The Potsdam Institute for Climate Impact Research (PIK)
2026	The different ways ecological change can happen and how systems respond* Invited seminar talk at the University of Alberta.
2025	The joy of getting stuck in traffic* Invited <u>education</u> talk at the winter 2025 Canadian Math Society meeting.
2025	Regime shifts in biology and tools to predict them* Invited research talk at the winter 2025 Canadian Math Society meeting.
2025	Regime shifts in resource explicit ecological systems* Invited seminar talk at McMaster University.
2025	Shining light on phytoplankton dynamics: How light availability and niche differentiation can promote coexistence in the water column* - Invited research talk at the 2025 Society of Mathematical Biology annual meeting.
2025	The many mechanisms behind regime shifts and tools to predict them* - Invited research talk at the 2025 Society of Mathematical Biology annual meeting.
2025	Regime shifts in resource explicit ecological systems* - Invited keynote speaker at Math & Ecology Synthesis for Agriculture (MESA) at Texas Tech University
2025	Modelling and Predicting Regime Shifts in Ecological Systems* - Invited research talk for the UC Davis Centre for Population Biology.
2025	Understanding transient crawl-bys in ecological timeseries* - Invited research talk at the Joint Math Meeting 2025
2024	Regime shifts and transient dynamics in ecological systems* - Invited research talk in the NOAA FED seminar at the NOAA Southwest Fisheries Science Center in Santa Cruz, CA.
2024	Model free method of predicting transient dynamics* - Invited research talk at SIAM Life Sciences meeting 2024

- 2024 Model free methods of predicting transient dynamics* - Invited research talk at American Math Society west 2024
- 2024 Understanding transient crawl-bys in ecological timeseries* & Coupling the socio-economic and ecological dynamics of cyanobacteria*. - Invited research talks at the Joint Math Meeting 2024.
- 2023 Model free methods of predicting transient dynamics^c - Research talk at the 2023 Ecological Society of America annual meeting.
- 2023 Parameter estimation and model selection*^L - Invited by Marissa Baskett to guest lecture for ECL/PBG 233 (Computational Methods in Population Biology, graduate level) at UC Davis.
- 2023 Model free methods of predicting transient dynamics* - Invited research talk at the 2023 Society of Mathematical Biology annual meeting.
- 2023 Coupling socio-economic and ecological dynamics of phytoplankton - Sustainable Oceans NRT One-Day Research Symposium
- 2023 Model free methods of predicting transient dynamics* - Invited research talk at the American Math Society western annual meeting.
- 2023 Model free methods of predicting transient dynamics* - Invited seminar talk at the University of Alberta.
- 2022 Model free methods of predicting transient dynamics* - Invited research talk at the Fields Institute: Workshop on Advances in Mathematical Ecology. Online recording found [here](#).
- 2022 Mathematical comparison and empirical review of the Monod and Droop forms for resource-based population dynamics* - Invited talk in the Canadian Math Society 2022 winter meeting
- 2022 Niche differentiation in the light spectrum promotes coexistence of phytoplankton species: a spatial modelling approach* - Invited talk at the Canadian Math Society 2022 winter meeting
- 2022 Understanding transient dynamics using empirical dynamical modelling* - Invited seminar talk at York University
- 2022 Modelling Phytoplankton Across Many Scales: Transient Dynamics and Human Interactions - Talk in the Mathematical Biology Seminar series at University of California, Davis.
- 2021 Niche differentiation in the light spectrum promotes coexistence of phytoplankton species: a spatial modeling approach*^O - Invited research talk at the 2021 annual Society of Mathematical Biology meeting.
- 2021 Coupling the socio-economic and transient ecological dynamics of cyanobacteria*^O - Seminar talk at Texas Tech University.
- 2021 Coupling the socio-economic and ecological dynamics of cyanobacteria*^O - Talk at the BIRS workshop: Mathematics of Human Environmental Systems.
- 2020 Coupling the socio-economic and ecological dynamics of cyanobacteria*^O - Talk at The Mathematical Biosciences Institute meeting: Life on Planet Earth: Above and Below.
- 2020 Modeling Cyanobacteria: Transient and social-ecological dynamics* - Seminar talk at the University of British Columbia, Okanagan.
- 2019 Transient dynamics of a stoichiometric cyanobacteria model^C - Talk at the Seventh International Conference on Mathematical Modeling and Analysis of Populations in Biological Systems (ICMA VII).

- 2019 Multiple-scale analysis of a stoichiometric cyanobacteria model with phosphorus impulses* - Presentation in the mini symposium, “Resource explicit population models”, during the 2019 annual Society of Mathematical Biology meeting at the Université de Montréal.
- 2019 Perturbation Theory for Uses in Mathematical Biology*^L - International Undergraduate Summer Enrichment Program (IUSEP).
- 2019 Multiple-scale analysis of a stoichiometric cyanobacteria model with phosphorus impulses^C - Talk at Levin Fest.
- 2019 Slow Manifold and Perturbation theory in Mathematical Biology*^L - Invited by Hao Wang to guest lecture for MATH570 (Mathematical Biology, graduate level) at U of Alberta.
- 2018 Stoichiometric Modeling and Multi-Scale Dynamics of Cyanobacteria* - Talk at the 6th Butler memorial conference on Differential Equations and Population Dynamics.
- 2018 Stoichiometric Modeling and Multi-Scale Dynamics of Cyanobacteria^C - 2018 annual CAIMS meeting.
- 2015 Ratio-Dependence in Predator Prey Dynamics - First talk of the Undergraduate Mathematics Seminar series at Ryerson University.

Outreach & other relevant activities

- 2022-Current **Protect Our Winters Canada- Science Alliance**
Confirming/discussing science based approaches to combating and advocating for environmental issues affecting outdoor athletes and recreation.
- 2023 **Ecological Society of America 2023 annual meeting volunteer activities**
A mentor to the workshop “think like a modeler”. Judge for Lotka and Volterra awards. Mentor for CV writing workshop.
- 2022 **Mentor for the UC Davis Ecology and Evolution Graduate Preview Program.**
- 2022 **Ecological Society of America’s Black Ecologists, Early Career, and Inclusive Ecology Sections panel discussion: ‘Writing Diversity and Inclusion Statements’**
- 2021-Current **Member of the Joint Baskett, Schreiber, Hastings, etc Lab meetings.**
Active member of the Marissa Baskett, Sebastian Schreiber, Alan Hastings and other joint lab meetings. I contribute to the group in various ways including, but not limited to, organization, research inputs, and presenting my own research.
- 2021-Current **Member of the Alan Hastings Research Lab.**
Active member of Alan Hastings research group contributing to the group in various ways including, but not limited to, organization, research inputs, writing of DEI statements, and presenting my own research.
- 2020 **MEOPAR webinar: ‘BIPOC experiences in STEM: Confronting racism in academia’**
- 2015-2021 **Member of the Lewis Research Group**
I was an active member of the Mark Lewis research group contributing to the group in various ways including, but not limited to, organizing events, research inputs, writing of DEI statements, and presenting my own research.
- 2015-2021 **Member of the Hao Wang Interdisciplinary Lab for Mathematical Ecology & Epidemiology**
I was an active member of Hao Wang’s ILMEE contributing to the group in various ways including, but not limited to, organizing events, research feedback, and presenting my own research.
- 2015-2020 **Department of Mathematical and Statistical Sciences Math Outreach**
Volunteer in math outreach at various events, including outreach to rural high schools.
- 2015-2020 **Math Fair & Unfair**

Assisted in the University of Alberta event for elementary and junior high aged students, encouraging interest in mathematics through games and other activities.
